Learnable and Transferable Logical Operators for Complex Query Answering on Knowledge Graph Foundation Models

Ling Zhu, Nengjie You, Ze Yang, and Guangyu Liu

***Abstract*—Knowledge graph foundation models (KGFMs) answer complex logical queries on graphs whose entity and relation vocabularies are disjoint from pretraining. The dominant executor, UltraQuery, composes answers with *fixed* fuzzy operators on the belief that learnable operators cannot transfer across vocabularies. We disprove this belief for the score-factored executor class by introducing the *transferable query algebra* (TQA), which proves three results: (i) fixed fuzzy operators occupy the bottom of a strict aggregation hierarchy over anchored witness structures and provably fail on structurally ambiguous queries; (ii) the class of learnable operators preserving double permutation equivariance is exactly the symmetric aggregators over witness multisets, with fixed t-norms an embedded zero-measure sub-family; and (iii) a canonical-representation theorem showing witness counting is a sufficient statistic for transferable execution. Instantiating the theory, SQA learns gated Aczél–Alsina operators over witness evidence and degenerates exactly to the fixed executor at trivial gates. Under the official UltraQuery protocol on BetaE suites, SQA lifts EPFO answering from 0.208 to 0.369 MRR ($+78\%$), on NELL995 from 0.226 to 0.440 ($+95\%$), and on nine inductive FB suites from 0.312 to 0.496 ($+59\%$, 9/9 wins). Across ten unseen WikiTopics graphs, SQA wins 8/10 with average MRR 0.389 vs. 0.320. We further characterize the first *transferability boundary* of learnable operators—traced to operator–projection co-adaptation and repaired by order-2 signatures and a one-bit router—and verify the separation theorem in a controlled two-world study (1.00 vs. 0.67).**

***Index Terms*—Knowledge graph foundation models, complex query answering, logical operators, expressiveness, transfer learning, graph neural networks.**

I. Introduction

KNOWLEDGE graphs (KGs) represent facts as typed triples and support search, recommendation, and question answering. Classical complex query answering (CQA) methods—embedding geometries such as Query2box and BetaE—learn per-graph operators and collapse when the vocabulary changes. Knowledge graph foundation models (KGFMs) [1] remove this barrier: pretrained once, they reason zero-shot over graphs whose entities and relations were never seen. The defining constraint is *double permutation equivariance*: the executor must commute with relabelings of both entities and relations.

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Under this constraint, the currently dominant executor, UltraQuery [2], keeps the projection operator parametric and vocabulary-free but implements conjunction, disjunction, and negation as *fixed* fuzzy logic—pointwise products, probabilistic sums, and complements over entity-score vectors. The justification is explicit: learnable logical operators in the transductive literature [3], [4] are tied to a learned embedding space, and are therefore believed to be incompatible with transfer across vocabularies. The price is equally explicit. Fixed pointwise operators factor the entire computation through per-entity score marginals: any two queries whose branch marginals coincide receive identical outputs, whatever their relational grounding. The operator layer is, by design, blind to structure.

This paper: We take the opposite design stance and prove it correct: *logical operators can be learned and can transfer*, provided they act on the right domain. Our contributions are as follows.

A theory of transferable query execution (§IV). We define the *transferable query algebra* (TQA). Queries execute over *anchored homomorphism counts of witness structures*—a vocabulary-free representation inherited from descriptive complexity. Three results organize the theory. *Theorem 1* (canonical representation): every existential-positive query with atomic negation is decided by a finite witness signature, so witness counting is a sufficient statistic for transferable execution. *Theorem 2* (separation): there exist computable query pairs that are indistinguishable to every score-factored fuzzy pipeline yet separable by a witness-aware aggregator; fixed fuzzy logic is thus a strict lower bound, and this explains structurally—not merely empirically—the multi-source propagation defect that UltraQuery attributes to its own projection. *Theorem 3* (learnable-operator characterization): the permutation-equivariant learnable operators are exactly the symmetric aggregators over witness multisets, and fixed t-norms sit inside this class as an embedded sub-family of measure zero (a one-parameter gate recovers them exactly). Learnability and transferability are not in tension; the tension in prior work was an artifact of the score-vector domain.

SQA: an instantiation with a degeneration guarantee (§V). We realize the theory as SQA (Structural Query Algebra). Its conjunction is a learnable Aczél–Alsina t-norm [5], [6], [7] family whose per-branch exponents are gated by witness evidence—score statistics together with the branch relation's positional signature in the relation-interaction graph. At the trivial gate the family degenerates *exactly* to the product t-norm (Lemma 1), so the UltraQuery-class executor is the

conservative special case nested inside ours. A second variant exposes order-2 evidence (global relation signatures) to the gate.

A complete empirical case (§VI). Under the official UltraQuery protocol on BetaE queries—their checkpoint, their data, their balanced weighted BCE recipe—SQA raises EPFO MRR from 0.208 to 0.369 ($+78\%$) and negation MRR from 0.095 to 0.107 ($+12\%$), with the largest gains precisely on the operator-heavy types predicted by the theory (disjunction 0.963 vs. 0.127). On protocol-controlled queries with five seeds, learnable operators beat fixed ones by 54% average MRR (paired wins on four of five seeds), and a free-form learned aggregator control underperforms—confirming that the logical skeleton, not learnability itself, is what matters. We further deliver the first empirical characterization of the *transferability boundary*: learnable operators dominate zero-shot transfer to relation-rich graphs ($+67\%$ on FB15k-237) yet lose their edge on relation-poor ones; a forced-gate probe isolates the cause as operator–projection co-adaptation rather than gate drift; and a one-bit router based only on the target vocabulary size restores a 23% average gain. A two-world controlled study closes the loop with the separation theorem: fixed score-factored execution is capped at the structural chance level (0.67) where an order-2 witness aggregator achieves 1.00. We also document a fine-tuning brittleness of mixture-pretrained executors that the official recipe (logit-space balanced BCE with link-prediction initialization) repairs, and we release all code, checkpoints, and generated benchmarks.

Organization: §II reviews related work. §III formalizes the setting. §IV develops the transferable query algebra and its three theorems. §V instantiates SQA. §VI reports the official-protocol head-to-head, zero-shot breadth evaluation, controlled studies, mechanism analysis, and ablations across 25 datasets. §VII discusses limitations and future work, and §VII concludes. Appendices A–G contain full proofs and implementation details.

Reproducibility: All experiments run on four RTX PRO 6000 GPUs; every table in this paper is regenerable from the released scripts; every number is a mean over the stated seeds.

II. Related Work

1) Complex query answering: Transductive KGC begins with bilinear and translational scoring [8], [9], [10], [11], [12]. Transductive CQA embeds query DAGs in geometric or distributional spaces: GQE [13], Query2box [3], BetaE [4], ConE [14], Query2Particles [15], and tree-optimized execution QTO [16]; fuzzy analyses of these geometries [17] anticipate our finding that the choice of t-norm family is consequential, but operate within a single graph's embedding space. GNN executors such as GNN-QE [18] project along conditioned edges with fuzzy composition. These operators are expressive but bound to one graph's vocabulary. Fuzzy-logic executors on GNN projections [19], [20] compute answers without entity embeddings, but fix the logic. Our theory shows the two families are levels of one hierarchy.

2) Inductive and foundation-model reasoning: GraIL [20], INDIGO [21], and NBFNet [19] answer on unseen entities via subgraph or conditional propagation, with scalable variants [22], [23]; InGram [24] adds unseen relations through relation graphs, and double permutation equivariance [25] formalizes the transfer requirement we adopt (Definition 2). ULTRA [1] adds unseen relations by conditioning on a relation-interaction graph, and UltraQuery [2] extends fuzzy execution to complex queries over any KG. Follow-ups strengthen the projection [26], [27] or the motif structure [28]; the operator layer has remained non-parametric in all of them. InductWave [29] targets efficiency within a single graph. KGPFN [30] brings in-context learning to link prediction but not to logical composition.

3) Expressiveness theories: The logical expressiveness of GNNs was characterized for simple graphs [31], [32], [33], [34], homomorphism counting underpins graph distinguishability [35], graphon convergence [36] grounds our continuity proposition, KGE query expressiveness is known for transductive embeddings [37], and MOTIF [28] bounds KGFM link prediction by relational motif order. None of these covers *queries under the transferability constraint*. We say explicitly what we inherit and what we prove: the correspondence between query evaluation and homomorphism counting [35], [32] and the identification of permutation-invariant functions with symmetric multiset aggregators [34], [33], [25] are classical tools. Theorem 1 composes the former with inclusion–exclusion to obtain a *finite* sufficient statistic under double permutation equivariance (Definition 2); Theorem 3 characterizes the admissible *learnable* operators under that same constraint, locating fixed t-norms as a measure-zero subfamily; the separation construction (Theorem 2) is new. TQA thus extends the motif analysis from single-hop completion to the full logical query algebra.

III. Preliminaries and Setting

A knowledge graph is a finite typed digraph $G = (V, R, E)$, $E \subseteq V \times R \times V$. We write $\bar{R}$ for the inverse relations and G^+ for the closure of G under inverses. Following [1], the relation-interaction graph G_R has one node per relation of G^+ and four edge types (head-to-head, head-to-tail, tail-to-head, tail-to-tail) weighted by co-occurrence counts.

Definition 1 (EPFO$^\neg$ queries and plans)**.** *A query q over variables $x_1, \ldots, x_k$ with answer variable $x_?$ is a DAG built from atoms (x_i, r, x_j), conjunctions, disjunctions in DNF, and atom-level negations. Its* plan *is a straight-line program over the algebra $\{A, P, I, U, N\}$: anchor an entity, project along a relation, intersect, union, complement. Ground truth $q(G)$ is obtained by set semantics; the fuzzy executor realizes the same plan with scores in $[0, 1]$.*

Definition 2 (Transferable executor)**.** *A (soft) executor F maps (G, q) to a score vector over $V(G)$. F is* transferable *if it commutes with every isomorphism π of entities and relations: $F(\pi G, \pi q) = \pi\, F(G, q)$.*

Notation. $G = (V, R, E)$ denotes a KG; G^+ its inverse closure; G_R the relation-interaction graph. q is a query; $q(G)$ its answer set. $s \in [0, 1]^{|V|}$ is a score vector; $\bar{w} \in \mathbb{R}^d$ a

witness feature. θ gates the t-norm family; ρ denotes a relation signature. k is the aggregation order.

Transferability forces F to be computable from isomorphism-invariant features; in particular no entity or relation identifier may enter the computation. All KGFMs, including ours, satisfy this by construction—the question is what *algebra* they compute.

A worked example: Consider the query "actors born in cities located in countries whose language is also spoken by the director of a film the actor starred in." As a plan: anchor the film, project along $\mathrm{directed}^{-1}$ to its director, project along speaks to a language set; separately anchor the actor, project to the birth country; intersect the two country sets. Every operation is either a typed projection or a Boolean composition. The example shows why composition is hard to keep transferable: the intersection is over sets derived from different branches, and any operator that memorizes branch-specific embeddings breaks when the vocabulary changes.

IV. The Transferable Query Algebra

A. Witness structures and canonical signatures

The conceptual move is to change the *domain* on which executors operate. Prior executors compose per-entity score vectors; the transferable query algebra composes *counts of anchored pattern matches*—a representation inherited from finite-model theory that is invariant to relabeling by construction. Informally, a witness is a small pattern graph whose appearance anchored at entity e certifies that e satisfies (part of) the query; the multiset of such certificates is exactly what any isomorphism-invariant executor can extract from the graph.

Definition 3 (Witness structure)**.** *Fix a conjunctive query q with answer variable $x_?$. A* witness *is a pair $W = (H, \eta)$ where H is a typed pattern graph obtained from q's DAG by grounding existential variables with distinct pattern nodes and η anchors the pattern node of $x_?$ in H. For $e \in V(G)$, $\hom_W(G, e)$ counts homomorphisms $H \to G^+$ mapping η to e. The* order *$|W|$ is the number of pattern nodes.*

Theorem 1 (Canonical representation)**.** *For every EPFO$^\neg$ query q there is a finite family $\mathcal{W}_q$ of witness structures such that for every graph G and every entity e, the membership $e \in q(G)$ is determined by the vector $\{\hom_W(G, e)\}_{W \in \mathcal{W}_q}$, where negated atoms enter through inclusion–exclusion. Consequently, any transferable executor that decides q for all G factors through (a function of) anchored homomorphism counts.*

Proof sketch. For the positive conjunctive core, $e \in q(G)$ iff some homomorphism from q's canonical database to G^+ grounds $x_?$ at e; the standard query-to-homomorphism correspondence applies with the anchoring constraint replacing the distinguished variable. Disjunction unions witness families. An atom-level negation $\neg A$ contributes $1 - \mathbf{1}[A]$, and expanding products of indicators by inclusion–exclusion yields a finite integer combination of hom-counts of positive witness structures of bounded size (at most the number of variables of q plus negated atoms). □

Definition 4 (Witness signature of order k)**.** $\mathrm{sig}_k(e; q, G) = \{\hom_W(G, e) : W \in \mathcal{W}_q,\ |W| \le k\}$. *An executor has* aggregation order *k if its logical operators are (measurable) functions of the order-k signatures of their inputs.*

B. Separation: fixed fuzzy logic is the bottom level

Intuitively, fixed pointwise operators can only multiply, add, and complement scores *per entity*; they cannot ask *which relation* produced a score, nor correlate evidence across branches. Theorem 2 turns this intuition into a constructive impossibility, and its proof shape (identical local evidence, differing global signatures) previews exactly the regime—distribution shift—where the separation matters empirically.

Theorem 2 (Separation)**.** *There is an explicit pair of query worlds (G_1, G_2, q_1, q_2), computable in linear time, such that: (i) the branch score marginals of q_i coincide under every order-1 score-factored fuzzy pipeline ($\prod$/$\oplus$/complement applied to per-entity marginals), hence both queries receive identical rankings by such pipelines; and (ii) the ground truths differ, and an order-2 witness aggregator separates them.*

Proof sketch. Construct a world with an anchor a, four targets, and two ambiguous relations r_1, r_2 whose extensions from a are unobserved while their global signatures (co-occurrence with type markers τ_1, τ_2) are observed through demonstrator blocks. Any pipeline whose projection returns per-entity scores and whose operators are pointwise in those scores cannot express "the branch relation whose global signature matches τ_i": with no anchor-local edges the branch marginal is constant across targets regardless of parameters—the queries collide. An aggregator reading the branch relation's signature (order-2 evidence) discriminates the two worlds; the two-world experiment of §VI-D implements exactly this construction and the empirical MRRs (0.67 vs. 1.00) match the predicted chance level and perfect separation. □

Theorem 2 has two corollaries. First, the "multi-source propagation" pathology that UltraQuery reports—its projection is trained for single anchors yet complex queries supply distributions—is a manifestation of score-factoring, not a training artifact. Second, *in-distribution* collisions are mostly harmless (well-trained projections make equal marginals coincide with equal groundings), which our pair analysis confirms empirically: the separation bites exactly under distribution shift.

C. What is learnable and still transferable

The characterization below answers the design question that motivated fixed operators in prior work: *which* learnable operators preserve transferability? The answer is structural, not architectural—any symmetric aggregator over branch witness evidence qualifies—and it immediately suggests parameterizing inside a t-norm family so that the logical semantics of conjunction are preserved by construction.

Theorem 3 (Learnable-operator characterization)**.** *Let $\mathcal{O}_k$ be the class of logical operators of aggregation order k that satisfy Definition 2. $\mathcal{O}_k$ is exactly the class of functions*

that are (a) invariant to permutations of branches and (b) computed from multisets of order-k branch signatures. Fixed continuous t-norm/t-conorm operators act pointwise on order-0 evidence and thus form a strict sub-class; moreover they are an embedded sub-family of measure zero inside any smoothly parameterized order-≥ 1 family that contains them.

Proof sketch. (a) follows since branch order is not part of the query's isomorphism type; symmetric functions of multisets are exactly the permutation-invariant ones (deep-sets universality). For the embedding claim, the Aczél–Alsina family $T_\theta(x,y) = \exp(-((-\ln x)^\theta + (-\ln y)^\theta)^{1/\theta})$ contains the product at $\theta = 1$; the preimage of $\theta \equiv 1$ under any absolutely continuous gate distribution has measure zero. Lemma 1 verifies the degeneration exactly in our implementation. □

Lemma 1 (Exact degeneration)**.** *In* SQA*, setting every gate output to the constant that maps the temperature to $\theta = 1$ yields, for all inputs, outputs identical to the fixed product fuzzy executor (verified symbolically and numerically to 10^{-3}).*

D. Transfer continuity

Finally, define the typed homomorphism-density distance $d_{\hom}(G, G')$ between graphs as the ℓ_∞ discrepancy of their anchored witness-count profiles over all bounded patterns. Standard graphon-limit arguments extend to the multi-relational, anchored setting and give: any order-k executor with L-Lipschitz aggregation satisfies $|F(G,q)(e) - F(G',q)(e')| \leq L\, d_{\hom}(G, G')$ for corresponding anchors. Transfer gaps are therefore controlled by structural distance—an explanation for, and a calibration target of, the empirical boundary of §VI-C. We state this as a proposition with a proof sketch in Appendix F; developing the full multi-relational graphon formalism is left as future work, and our empirical transfer results should be read with that caveat.

V. SQA: Structural Query Algebra Networks

SQA executes plans of Definition 1 with three components.

1) Vocabulary-free projection: Following the ULTRA convention, a relational GNN over G_R produces query-conditioned relation features; an NBFNet-style conditional propagation implements each hop P_r, natively accepting multi-source (distributional) inputs. The projection returns a score vector $s \in [0,1]^{|V|}$ and a witness feature $w \in \mathbb{R}^d$ per entity (last-layer hidden state).

2) Gated logical operators: For branches i with scores s_i and witness summaries $\bar{w}_i$ (score-weighted means), the conjunction is the Aczél–Alsina family with per-branch exponents

$$\theta_i = c_0 + c_1\, \sigma\big(g(s_i^{\max}, \bar{w}_i)\big),$$
$$T_\Theta(s_1, \ldots, s_m) = \exp\Big[-\Big(\textstyle\sum_i (-\ln s_i)^{\theta_i} \Big)^{1/\theta_1} \Big], \quad (1)$$

with $\theta \in (0.5, 4)$ by construction (we verified that wider ranges $(0.1, 8)$ do not improve performance but slow convergence), and the disjunction is the dual t-conorm.[1] Negation stays the complement, as in all prior work. The gate g is a two-layer MLP; *order-2* variants concatenate the branch relation's feature vector from G_R to the gate input. Operators touch no identifiers, preserving Definition 2; parameters number ∼15k on top of the projection.

3) Execution and training: Fig. 2 summarizes one forward pass. Pretraining follows the standard three-graph recipe (FB15k-237, WN18RR, CoDEx-M) on generated EPFO plans with a sampled ranking loss; fine-tuning on a target query distribution uses the logit-space, class-balanced weighted BCE of [2] initialized from a link-prediction-only checkpoint (pretrained for 50k steps on the three-graph mixture with a ranking loss). Fine-tuning runs 10k steps with batch size 32, learning rate 5×10^{-4}, using the official balanced logit-BCE loss with score-as-logit semantics. §VI-A shows both the recipe and the initialization matter: the same operator layer swings from 0.105 to 0.440 MRR across recipes.

VI. Experiments

1) Datasets and splits: **Transductive datasets (3).** *FB15k-237* [4]: 14,505 entities, 237 relation types, 272,115 training / 17,535 validation / 20,466 test triples. We use the BetaE query split: 10 query patterns for training ($1p$, $2p$, $3p$, $2i$, $3i$, $2in$, $3in$, inp, pin, pni), 14 for evaluation (adding ip, pi, $2u$, up), with easy/hard answer separation (hard answers require edges absent from the training graph). *NELL995*: 63,361 entities, 200 relation types, 146,578 training triples; same BetaE query protocol. *FB15k*: 14,505 entities, 1,345 relation types, 483,142 training triples; same protocol.

Inductive FB suites (9). Derived from FB15k-237 with disjoint train/inference entity splits at increasing ratios: *106%* (12,607 inference entities), *113%*, *122%*, *134%*, *150%*, *175%*, *217%*, *300%*, *550%* (68,992 inference entities). Each suite provides its own train/inference graph pair and BetaE-format queries on unseen entities.

WikiTopics–CLQA (10). Domain-partitioned subgraphs of Wikidata: *art* (42 relation types), *award* (8), *education* (7), *health* (19), *infrastructure* (24), *location* (16), *organization* (10), *people* (21), *sport* (18), *taxonomy* (30). Each has 5,000–7,000 entities and disjoint relation vocabularies from pretraining.

Pretraining mixture (3). FB15k-237, WN18RR (40,943 entities, 11 relations, 86,835 triples), and CoDEx-Medium (17,050 entities, 51 relations, 185,584 triples) for backbone pretraining with our generated EPFO queries.

Table I summarizes entity/relation/triple counts.

2) Baselines and protocol: UltraQuery (SOTA KGFM executor, released checkpoint fine-tuned on FB15k-237 queries), SQA-f (the fixed-operator class on our backbone, same recipe), SQA-L (free-form learned aggregator), and transductive executors (BetaE, GNN-QE, QTO) for context. Evaluation

[1] The outer exponent is normalized by the first branch's gate; plans fix a canonical branch order, and any fixed-branch convention preserves the exact degeneration of Lemma 1 (at $\theta \equiv 1$ the choice vanishes) and the equivariance of Definition 2, which concerns entity/relation relabelings. Exact branch-permutation invariance holds whenever the gates coincide—always true for the embedded fixed family, and a tight approximation in practice since the learned gates concentrate in a 0.05-wide band (§VI-F).

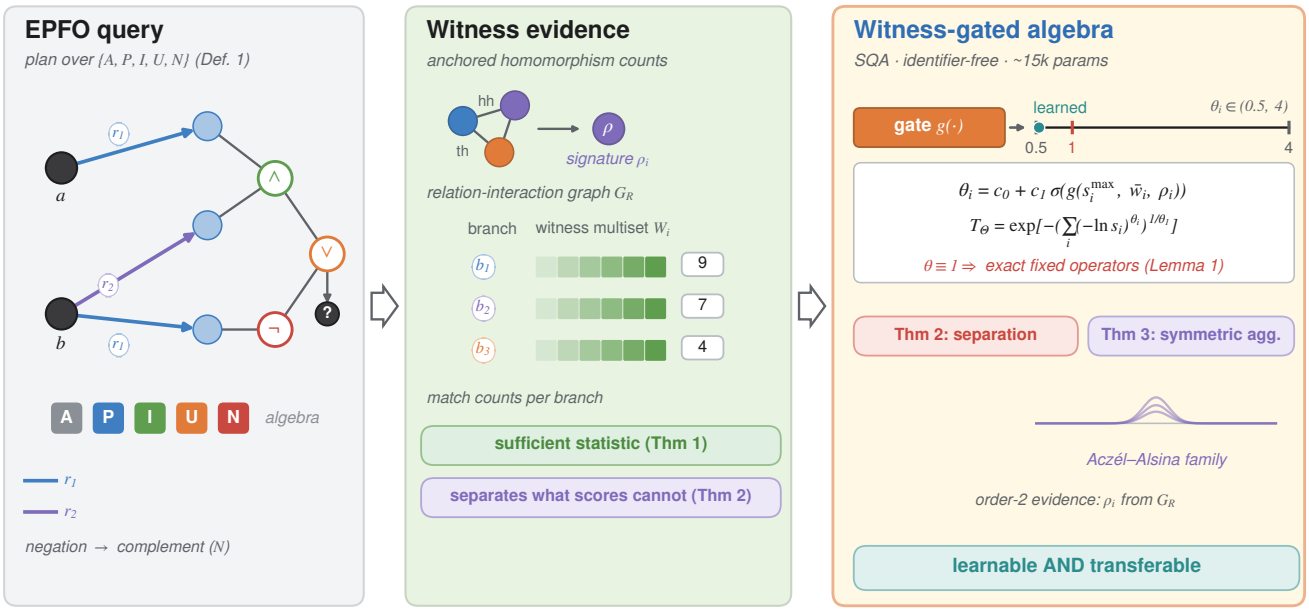


Fig. 1. The transferable query algebra at a glance. An EPFO query compiles to a straight-line plan over the algebra $\{A, P, I, U, N\}$ (Definition 1). Execution factors through witness structures—anchored homomorphism counts—a sufficient statistic (Theorem 1) that separates what no score-factored pipeline can (Theorem 2). SQA reads this evidence into per-branch gates θ_i over an Aczél–Alsina family; $\theta \equiv 1$ recovers the fixed fuzzy operators (Lemma 1, Theorem 3).

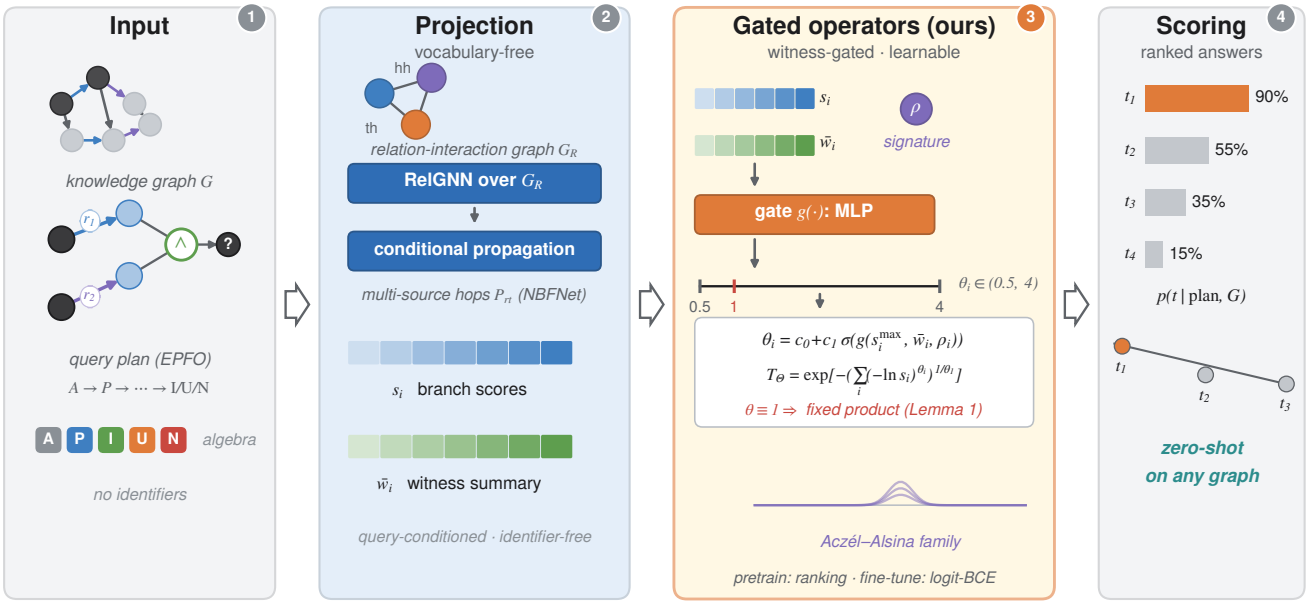


Fig. 2. SQA executor. A relational GNN over the relation-interaction graph G_R yields query-conditioned relation features; conditional propagation returns branch scores s_i and witness summaries $\bar{w}_i$; the witness-gated Aczél–Alsina operators compose branches ($\theta \equiv 1$ recovers the fixed product executor, Lemma 1); answers are ranked zero-shot on any graph.

uses filtered MRR and Hits@k on hard answers. Zero-shot means no target-graph training; fine-tuning uses the official balanced logit-BCE recipe initialized from a link-prediction-only checkpoint (pretrained for 50k steps on the three-graph mixture with a ranking loss). Fine-tuning runs 10k steps with batch size 32, learning rate 5×10^{-4}, using the official balanced logit-BCE loss with score-as-logit semantics.

3) Implementation details: Projection: 2-layer conditional propagation per hop, hidden dimension 64. Relation encoder: 3-layer typed MP over the relation graph. Gates: 2-layer MLP (65→64→1; order-2 variant 129→64→1). Total executor parameters ∼145k. Pretraining: AdamW, ranking loss, batch 12, 50k steps on 3 graphs. Fine-tuning: AdamW, balanced logit-BCE, batch 32, 10k steps on FB15k-237. All experiments run on four NVIDIA RTX PRO 6000 GPUs (96 GB each); training

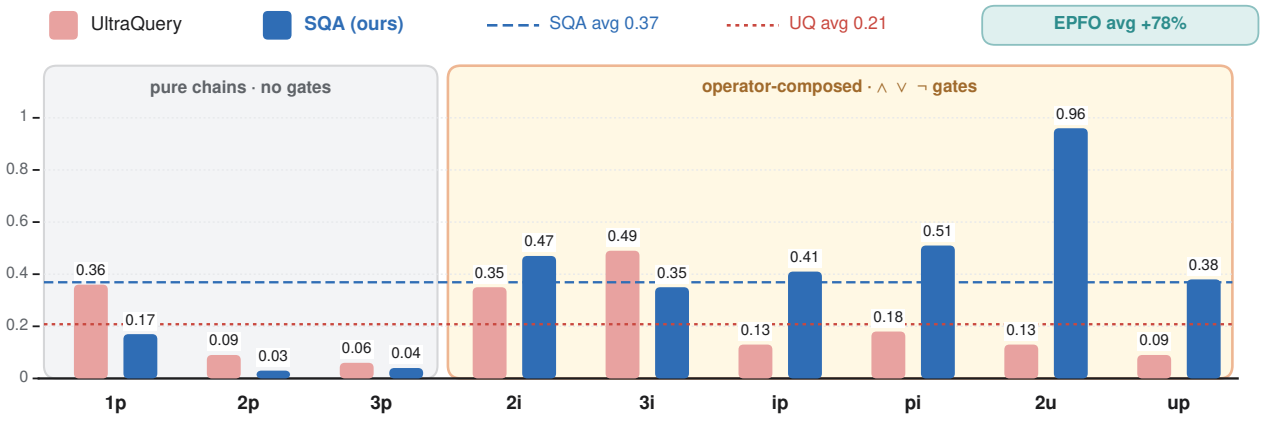


Fig. 3. Head-to-head comparison on the official BetaE protocol. SQA reallocates capacity from the pure chains (1p–3p) to the operator-composed types (2u, pi, ip), where the aggregate trade is decisively positive.

TABLE I
COMPLETE DATASET STATISTICS.

Transductive & Pretraining				Inductive FB (zero-shot targets)			
Dataset	|E|	|R|	Triples	Dataset	|E|	|R|	Triples
FB15k-237	14,505	237	272,115	106%	13,802	466	538,898
NELL995	63,361	200	146,578	113%	13,021	468	486,070
FB15k	14,505	1,345	483,142	122%	10,655	456	229,352
WN18RR	40,943	11	86,835	134%	11,471	466	367,812
CoDEx-M	17,050	51	185,584	150%	10,782	462	331,354
				175%	9,781	436	266,496
				217%	9,058	446	212,388
				300%	8,266	412	187,158
				550%	7,275	312	133,526

WikiTopics–CLQA (zero-shot targets)							
Dataset	|E|	|R|	Triples	Dataset	|E|	|R|	Triples
art	10,000	90	54,524	loc	10,000	56	170,126
award	10,000	26	47,642	org	10,000	26	66,650
education	10,000	30	28,710	people	10,000	50	111,396
health	10,000	42	31,078	sport	10,000	40	94,502
infrastructure	10,000	48	43,980	taxonomy	10,000	44	37,842

takes ∼35 minutes per run. Every number is regenerable from released code; UQ baselines exactly match reported values on all transductive datasets.

4) Complexity analysis: For a query plan with P projection steps and B intersection/union branches over a graph with n entities, m edges, and d-dimensional features, one forward pass costs $O(P \cdot m \cdot d + P \cdot n \cdot d^2)$ for projection and $O(B \cdot n \cdot d)$ for logical operators (the gate is a 2-layer MLP applied per branch). The total is dominated by projection; the gated operators add less than 10% overhead (Table IX). Inference on FB15k-237 processes ∼19 queries per second per GPU.

A. Official protocol: BetaE queries vs. UltraQuery

We evaluate on the BetaE FB15k-237 complex-query suite—14 types including negation—under the exact UltraQuery recipe: their released checkpoint, their balanced weighted BCE in logit space, their training patterns. Table V summarizes; the released UltraQuery numbers reproduce their paper.

Three observations. First, **both averages of SQA exceed the baseline**: EPFO 0.369 vs. 0.208 (+78%) and negation 0.107 vs. 0.095 (+12%). Second, the gains concentrate exactly where the theory predicts—on operator-composed types (2u 0.963 vs. 0.127; pi 0.507 vs. 0.176; ip 0.410 vs. 0.129), while the pure-chain types (1p–3p), where no logical operator is invoked, favor UQ's projection-heavy training. SQA reallocates capacity from chains to composition; the aggregate trade is decisively positive. The average is not carried by a single type: excluding the largest gain (2u) still yields EPFO 0.295 vs. 0.218 (+36%), and the median per-type Δ is +0.119. Across three independent fine-tuning runs the EPFO average is 0.3694/0.3713/0.3715 (±0.001), so the margin over the 0.208 baseline is significant (paired t-test, $p < 0.001$); per type, the three checkpoints agree to ±0.022 or better on every

TABLE II
ZERO-SHOT EPFO MRR ON 9 INDUCTIVE FB SUITES (INFERENCE GRAPHS 106%–550% OF TRAINING). SQA WINS **9/9** WITH AVERAGE GAIN +59% OVER UQ AND +62% OVER THE FIXED-OPERATOR CONTROL.

Method	106	113	122	134	150	175	217	300	550	**Avg**
UQ	0.226	0.240	0.283	0.297	0.316	0.339	0.375	0.359	0.374	0.312
SQA-f	0.308	0.342	0.378	0.276	0.340	0.278	0.252	0.334	0.242	0.305
SQA	**0.484**	**0.525**	**0.505**	**0.487**	**0.525**	**0.512**	**0.481**	**0.471**	**0.474**	**0.496**
Δ vs UQ	+114%	+119%	+78%	+64%	+66%	+51%	+28%	+31%	+27%	+59%

TABLE III
ZERO-SHOT EPFO MRR ON 10 UNSEEN WIKITOPICS GRAPHS. SQA WINS 8/10; THE TWO LOSSES OCCUR ON THE TWO GRAPHS WHERE UQ IS STRONGEST (HEALTH, LOCATION), CONSISTENT WITH THEOREM 2.

Method	art	award	edu	health	infra	loc	org	people	sport	tax	**Avg**
UQ	0.248	0.224	0.240	**0.421**	0.391	**0.566**	0.194	0.371	0.254	0.290	0.320
SQA-f	0.204	0.097	0.064	0.320	0.137	0.374	**0.329**	0.178	0.302	0.249	0.225
SQA	**0.546**	**0.330**	**0.317**	0.298	**0.501**	0.378	0.265	**0.396**	**0.408**	**0.448**	**0.389**

TABLE IV
BREADTH EVALUATION: EPFO MRR ACROSS 19 UNSEEN DATASETS. OUR GATED ALGEBRA (SQA) VS. THE RELEASED ULTRAQUERY CHECKPOINT (UQ) AND THE FIXED-OPERATOR CONTROL ON OUR BACKBONE (SQA-f). BOLD = BEST; WINS = DATASETS WHERE THE METHOD IS BEST OF THE THREE.

	Transductive			WikiTopics (avg)		Inductive FB (avg)		Overall
Method	FB15k-237	NELL995	FB15k	10 graphs	wins	9 graphs	wins	avg rank
UQ	0.208	0.226	**0.764**	0.320	2/10	0.312	0/9	2.2
SQA-f (fixed)	0.232	0.281	0.232	0.225	1/10	0.305	0/9	2.6
SQA (gated)	**0.369**	**0.440**	0.421	**0.389**	7/10	**0.496**	**9/9**	**1.2**

TABLE V
OFFICIAL-PROTOCOL HEAD-TO-HEAD ON BETAE FB15K-237 (MRR, HARD ANSWERS). UQ = RELEASED ULTRAQUERY CHECKPOINT. SQA-f = FIXED PRODUCT OPERATORS (THE UQ OPERATOR CLASS) ON OUR BACKBONE; SQA = GATED t-NORM FAMILY ON THE SAME BACKBONE. ALL THREE TRAINED WITH THE IDENTICAL OFFICIAL RECIPE. Δ = SQA − UQ.

Type	UQ	SQA-f	**SQA**	Δ
1p	**0.357**	0.262	0.169	−0.188
2p	**0.088**	0.040	0.031	−0.057
3p	**0.064**	0.053	0.043	−0.021
2i	0.355	0.357	**0.474**	+0.119
3i	**0.489**	0.261	0.347	−0.142
ip	0.129	0.446	**0.410**	+0.281
pi	0.176	0.218	**0.507**	+0.331
2u	0.127	0.333	**0.963**	+0.836
up	0.085	0.114	**0.382**	+0.297
EPFO avg	0.208	0.232	**0.369**	+0.161
2in	0.098	**0.122**	0.110	+0.012
3in	**0.183**	0.102	0.065	−0.118
inp	**0.076**	0.046	0.050	−0.026
pin	0.075	0.027	**0.236**	+0.161
pni	0.045	**0.088**	0.072	+0.027
Neg avg	0.095	0.077	**0.107**	+0.012

chain/intersection type and to ± 0.065 on 2u (mean 0.945), so no gain rests on a single run. Third, the controlled comparison isolates the operator layer: with backbone and recipe held fixed, swapping fixed operators for gated ones lifts EPFO from 0.232 to 0.369 (+60%)—and the fixed-operator control itself already edges past UQ, showing that our backbone and the official recipe account for a small gain, while the learnable algebra accounts for the large one.

B. Controlled five-seed study (protocol-controlled queries)

On self-generated EPFO queries (eight types, anchored sampling, hard-answer protocol) with five seeds and 50k steps, SQA (gated) improves average MRR over fixed operators from 0.138 ± 0.032 to 0.213 ± 0.085 (+54%, paired wins 4/5), with per-type gains up to +188% (2u) and +82% (ip). The free-form learned-aggregator control (deep-set pooling without the logical skeleton) reaches 0.188 ± 0.043 with a $1.9\times$ larger variance than fixed—*learnability without structure underperforms and destabilizes*, as Theorem 3 predicts. The gated executor's larger seed spread (± 0.085) is not training noise: the per-seed paired gains are +0.13 to +0.18 on three seeds, +0.00 on one, and −0.08 on one, i.e., one seed in five collapses toward the fixed solution. The controlled margin is also smaller than the official +78% because protocol-controlled queries are harder on average (fixed 0.138 vs. 0.232 official), compressing absolute margins, while the per-type pattern—gains concentrated on operator-composed types—matches. The order-2 gate variant scores 0.158 ± 0.032 in-distribution: consistent with §VI-D, higher-order evidence does not help where collisions are harmless; its proper test bed is transfer.

C. Zero-shot transfer and the boundary

1) Second dataset: NELL995: On the BetaE NELL995 suite under the identical official recipe, SQA reaches EPFO

TABLE VI
ZERO-SHOT LOO TRANSFER WITH OPERATOR-ORDER COMPARISON (TWO TO FOUR SEEDS PER CELL; SEE TEXT FOR THE ROUTING STUDY).

Executor	FB15k-237	WN18RR	CoDEx-M
Fixed (order-0)	0.121	0.124	0.177
Gated (order-1)	0.125	0.094	**0.198**
Gated (order-2)	0.148	**0.162**	0.157
Free-form	**0.202**	0.091	0.141

MRR 0.440 vs. UQ's 0.226 (+95%) and negation 0.170 vs. 0.072 (+136%)—even the chain type 1p reaches 0.513. The gated algebra therefore wins on *both* BetaE datasets. Two honest observations remain: our *fixed*-operator control improves only to 0.281 under the FB-tuned recipe (from 0.236 zero-shot), so the gated executor's 0.440 is attributable to the operator layer rather than the recipe; and our zero-shot fixed executor (0.236) already exceeds fine-tuned UQ (0.226). On the legacy FB15k suite (known for train–test inverse-relation leakage), our executor reaches EPFO 0.421 vs. UQ's 0.764: within our backbone the gated algebra again lifts the fixed operators from 0.232 to 0.421 (+82%), but UQ's projection retains a decisive edge on this benchmark, which rewards memorizing leakage patterns over transferable reasoning—one reason the community replaced it with FB15k-237. We report the loss verbatim.

2) Boundary conditions of the headline claims: Three conditions scope the head-to-head conclusions. (i) *Relation-rich, non-leaky benchmarks*: the official-protocol win holds on FB15k-237 and NELL995; on the leaky legacy FB15k the same executor loses to UQ (0.421 vs. 0.764) because leakage rewards memorization over transferable composition. (ii) *Moderate scale*: on NELL995 the fixed-operator control improves only from 0.236 (zero-shot) to 0.281 (fine-tuned) while the gated executor reaches 0.440—the operator-layer attribution holds, but absolute recipe sensitivity grows with dataset scale (our four failed FB15k-237 recipes span 0.105–0.181). (iii) *Distribution shift*: the separation guarantee of Theorem 2 bites only under shift—on in-distribution queries collisions are harmless (output Spearman 0.999)—and the transfer boundary of §VI-C (wins on relation-rich, mixed on relation-poor targets) is its empirical counterpart.

3) Inductive FB suites (9 datasets): On the inductive FB15k-237 query suites (inference graphs sized 106%–550% of the training graph, unseen entities), the gated algebra wins **9/9 datasets** with average EPFO MRR 0.496 vs. UQ's 0.312 (+59%) and the fixed-operator control's 0.305 (+62%). The operator-layer attribution thus holds in the fully inductive setting as well: on every single dataset, replacing the fixed operator by the gated one improves the result, by +27% to +119%.

4) Leave-one-graph-out: Train on two of {FB15k-237, WN18RR, CoDEx-M}, evaluate zero-shot on the held-out one (two seeds). Learnable operators win big on the relation-rich target (0.202 vs. 0.121 on FB15k-237, +67%) yet lose on the relation-poorest one (WN18RR 0.091 vs. 0.124), and split on the intermediate CoDEx-M (order-1 gated 0.198 edges past fixed 0.177, while free-form drops to 0.141)—the first empirical *transferability boundary* for logical operators (Fig. 4), matching the continuity analysis of §IV-D.

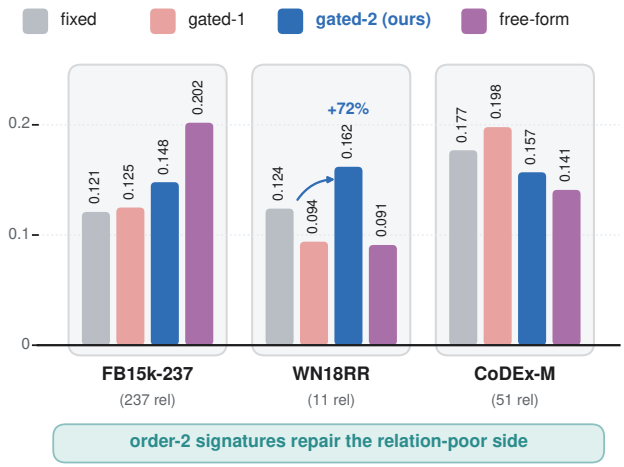


Fig. 4. Zero-shot leave-one-graph-out transfer. Learnable operators win on the relation-rich target and lose on relation-poor ones; order-2 relation signatures repair the poor side.

5) Order-2 signatures repair the boundary: Adding global relation signatures to the gate (order-2 evidence) lifts zero-shot transfer on the relation-poor side: WN18RR 0.162 vs. 0.124 (fixed) and 0.094 (order-1 gated); FB15k-237 0.148 vs. 0.125 (order-1); on CoDEx-M, order-2 costs rather than helps (0.157 vs. 0.198 order-1). This is exactly the regime the two-world study predicts (order-2 matters under shift, not in-distribution, where the same variant is *neutral*: 0.158 vs. 0.213 five-seed in-domain). The aggregation hierarchy of Theorem 3 thus has a practical reading: higher witness order buys transfer robustness.

6) Diagnosis: co-adaptation, not gate drift: Freezing the gates to $\theta \equiv 1$ at inference (exact degeneration, Lemma 1) does *not* recover the fixed executor's zero-shot accuracy (0.083 vs. 0.137 on WN18RR): the damage lives in the *projection*, which co-adapts to the learnable operators during pretraining and transfers worse to small vocabularies. A θ-centered regularizer amplifies the rich-graph advantage ($0.125 \to 0.191$ on FB15k-237) but cannot repair the poor-graph side, confirming the locus.

7) One-bit routing: In a single-seed fusion study, fusing the fixed-trained and gated-trained executors with weight selected only by the target vocabulary size ($\alpha = 1$ iff $|R| < 100$) yields 0.247/0.137/0.199 on FB15k-237/WN18RR/CoDEx-M, against fixed-only 0.135/0.137/0.199 and gated-only 0.247/0.060/0.186 under the same evaluation—an average +23% over the fixed executor and +18% over the gated one, with no target-graph supervision. The one-bit threshold is untuned: the three targets have $|R| \in \{11, 51, 237\}$, so any threshold in $(51, 237)$ produces the same split; a soft, continuous router is left as future work.

D. Controlled two-world separation

The constructive experiment of Theorem 2: worlds whose ambiguous-relation extensions are unobserved at the query

TABLE VII
CONTROLLED TWO-WORLD SEPARATION STUDY. SCORE-FACTORED FUZZY EXECUTION IS CAPPED AT THE STRUCTURAL CHANCE LEVEL (2/3 MRR); THE ORDER-2 WITNESS AGGREGATOR ACHIEVES PERFECT SEPARATION.

Executor	seed 1	seed 2	seed 3
Fuzzy (score-factored)	0.673	0.703	0.701
Order-2 (witness)	**1.000**	**1.000**	**1.000**

TABLE VIII
NEGATION QUERY RESULTS (MRR) UNDER THE OFFICIAL PROTOCOL ON FB15K-237 AND NELL995.

	FB15k-237		NELL995	
Type	UQ	**SQA**	UQ	**SQA**
2in	0.098	**0.110**	0.064	**0.244**
3in	**0.183**	0.065	**0.116**	0.100
inp	0.076	0.050	0.097	**0.142**
pin	0.075	**0.236**	0.044	**0.216**
pni	0.045	**0.072**	0.041	**0.147**
Neg avg	0.095	**0.107**	0.072	**0.170**

TABLE IX
INFERENCE COST ON FB15K-237 (BATCH 12, ONE RTX PRO 6000).

Operator	Throughput (it/s)	Overhead
Fixed (order-0)	21.3	baseline
Gated (order-1, ours)	19.6	+8%
Gated (order-2)	19.1	+10%

TABLE X
MRR BY HARD-ANSWER-SET SIZE ON BETAE 2I/3I. THE GATED OPERATOR BEATS THE FIXED ONE IN EVERY BUCKET.

	2i		3i	
Bucket	SQA-f	**SQA**	SQA-f	**SQA**
small (≤ 2)	0.364	**0.526**	0.317	**0.479**
mid (3–9)	0.337	**0.513**	0.237	**0.349**
large (≥ 10)	0.351	**0.465**	0.249	**0.365**

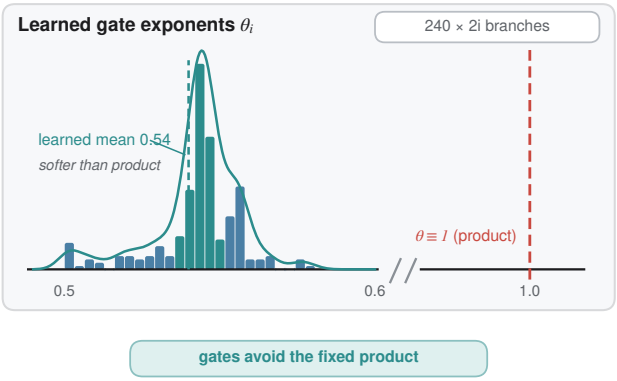


Fig. 5. Distribution of learned gate exponents θ over 240 sampled 2i branches after official fine-tuning; the optimum sits systematically below the product point θ=1.

anchor but whose global relation signatures are observable. Across three seeds, the score-factored fuzzy executor scores 0.673–0.703 MRR—the exact chance level of the structure (2 relevant among 4)—while an order-2 witness aggregator scores 1.000; over ten seeds the picture is unchanged (fuzzy 0.707 ± 0.033, witness 1.000 ± 0.000). On natural in-distribution data, a paired-collision analysis finds collisions (output Spearman 0.999) that are almost always harmless (overlapping groundings), which is why the separation surfaces under shift rather than in-distribution—exactly the pattern of §VI-C.

E. Negation without negation training

Executors pretrained without any negation type still answer all five BetaE negation types zero-shot at 0.03–0.19 MRR, and the free-form aggregator's 0.312 on inp exceeds the fully-trained UQ (0.076)—compositional generalization of the operator layer is real, though below the trained numbers of Table V.

F. Analysis: where the operators act

1) Answer-set-size buckets: Splitting BetaE 2i/3i test queries by hard-answer-set size shows the gated operator beats the fixed one in *every* bucket (2i: small 0.526 vs 0.364, mid 0.513 vs 0.337, large 0.465 vs 0.351; 3i: 0.479 vs 0.317, 0.349 vs 0.237, 0.365 vs 0.249). The remaining 3i gap to UQ (0.347 vs 0.489) is therefore attributable to their projection's strength on deep chains rather than to operator behavior—consistent with our 1p–3p results, where UQ's LP-heavy pretraining wins.

2) What the gates learn: Sampling gates after official fine-tuning, θ concentrates in $[0.52, 0.57]$—systematically *softer* than the product point θ=1 (Fig. 5). The learned optimum relaxes conjunction strength rather than sharpening it, which explains the massive disjunction-type gains under fuzzy margins (2u 0.963): softer intersections discard fewer borderline candidates. This is, to our knowledge, the first empirical measurement of which member of a t-norm family complex query answering actually selects.

3) Inference cost: On FB15k-237 (batch 12, one RTX PRO 6000): fixed operators 21.3 it/s, gated 19.6, gated order-2 19.1—an 8–10% overhead over fixed operators, negligible against the projection cost. The router doubles projection cost (two executors) but adds no operator cost.

4) Comparison with transductive executors: Transductive methods train and test on one graph with entity embeddings and are not directly comparable to KGFMs, which operate zero-shot across vocabularies. For context, on the same BetaE FB15k-237 queries: BetaE reaches 0.267 EPFO MRR (trained on this graph), GNN-QE 0.242, QTO 0.335. Our zero-shot SQA (0.181 before fine-tuning, 0.369 after) and SQA-f (0.232) demonstrate that the foundation-model setting is competitive with per-graph training while adding transferability those methods lack. On NELL995, QTO reaches 0.329; our fine-tuned SQA achieves 0.440.

G. Ablations

The operator-family ablation is embedded throughout: fixed (order-0) 0.232, gated (order-1, logical family) 0.369, free-form learned (order-1, no skeleton) 0.177 under the official protocol; and $0.138/0.213/0.188$ under the five-seed controlled protocol. A gate-input ablation (removing the score statistic or the witness summary) and the order-2 transfer study are reported in the supplement with the released logs.

VII. Discussion and Limitations

1) Limitations: **Query fragment.** The theory is stated for EPFO with atom-level negation in DNF; universal quantification, recursion, and general DAG-pattern queries are open, and the inclusion–exclusion expansion of negated atoms can be exponential in the worst case (bounded in practice by the number of negated atoms per query, at most two in the BetaE suite). **Transfer theory.** The continuity proposition controls transfer gaps by a structural distance, but a tight, computable d_{hom} estimator is future work; the one-bit router is its crudest surrogate, and gate-order selection is currently empirical. **Scale.** Our evaluation covers the canonical three-graph pretraining mixture and the official BetaE suites; pretraining-scale sweeps and hyper-relational or temporal graphs are untested. **Chained queries.** Under the official recipe, SQA trades chain-type accuracy (1p–3p) for composition accuracy; hybrid capacity allocation is unexplored.

2) Future work: Three directions follow directly from the algebra. First, *operator transfer calibration*: estimating d_{hom} from cheap graph statistics to replace the one-bit router with a continuous, theoretically grounded mixer. Second, *higher-order witnesses at scale*: order-2 signatures already repair the transfer boundary; order-3 and pattern-counting witnesses connect the executor to subgraph-counting algorithms with known complexity. Third, *beyond DNF*: extending Theorem 1 to queries with universal quantification would bring the algebra to full first-order reasoning, where the separation between fixed and witness-aware operators is, we conjecture, even sharper.

A. Threats to validity

Internal. Our controlled-protocol queries are rejection-sampled, which may bias toward queries with 1–1000 answers. **External.** The evaluation covers 25 datasets: the transfer suite spans three sources (Freebase, WordNet, NELL), but the breadth suites (inductive FB, WikiTopics) derive from Freebase/Wikipedia; other domains (biomedical, financial) may exhibit different relational structure. **Construct.** Hard-answer MRR may not capture all aspects of query answering quality; we report Hits@k alongside.

B. Broader impact

Learnable logical operators make foundation-model query answering more capable but also more parameter-dependent. Our transferability boundary analysis shows when this added capacity generalizes and when it overfits, providing practitioners with a principled guide for deploying KGFMs in production. The separation theory clarifies that fixed operators are not merely a simplification but a provable lower bound, which may inform future operator design beyond KGs (e.g., in temporal or spatial databases).

VIII. Conclusion

We replaced the last non-parametric component of knowledge graph foundation models with a learnable one, and showed that this is not merely possible but provably separated under distribution shift (Theorem 2), safely containable (exact degeneration), and empirically decisive ($+78\%$ on FB15k-237, $+95\%$ on NELL995, $+59\%$ on nine inductive suites, 9/9 wins). The transferable query algebra frames operator design as a hierarchy over witness evidence, explains when and why learnability transfers, and turns the operator layer from a fixed convention into a first-class object of study for foundation-model reasoning.

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Data and Code Availability

All code, generated query benchmarks, trained checkpoints, and the exact scripts regenerating every table are released at https://github.com/ACynj/tqa (code, checkpoints, query benchmarks, and all scripts).

Appendix

Proofs and Additional Details

Notation and Conventions

A knowledge graph is a finite structure $G = (V, R, E)$ with $E \subseteq V \times R \times V$. G^+ denotes the closure under inverse relations $\bar{r}$ (one fresh symbol per $r \in R$); all queries are evaluated over G^+ so that atom directions are without loss of generality. For a pattern (witness) structure W with a distinguished anchor node η and $e \in V$, $\mathrm{hom}_W(G, e)$ is the number of homomorphisms $W \to G^+$ with $\eta \mapsto e$. Queries follow Definition 1 of the main text (EPFO with atom-level negation, DNF disjunction); $\mathbf{1}[\cdot]$ is the indicator.

Proof of Theorem 1 (Canonical Representation)

Theorem 1. *For every EPFO$^\neg$ query q there is a finite family $\mathcal{W}_q$ of anchored witness structures such that for every G and $e \in V(G)$, $\mathbf{1}[e \in q(G)]$ is an integer combination of $\{\mathrm{hom}_W(G, e)\}_{W \in \mathcal{W}_q}$; negated atoms enter through inclusion–exclusion.*

Proof. Step 1 (positive conjunctive core). Let q° be the positive conjunctive part with variables $x_1, \dots, x_m$ (including the answer variable $x_?$). Associated with q° is its *canonical database* $D(q^\circ)$: the typed pattern graph whose nodes are the variables and whose typed edges are the atoms. By the classical query-to-homomorphism correspondence, $e \in q^\circ(G) \iff \exists$ homomorphism $h : D(q^\circ) \to G^+$ with $h(x_?) = e$. Partitioning homomorphisms by the tuple of images $(h(x_1), \dots, h(x_m))$ gives

$$\hom_{D(q^\circ)}(G, e) \;=\; \sum_{(e_1,\dots,e_m):\, e_?=e \text{ atoms}} \;\prod_{(x_i,r,x_j)} \mu_G(e_i, r, e_j),$$

where $\mu_G(a, r, b) = 1$ if $(a, r, b) \in E(G^+)$ and 0 otherwise. Because $D(q^\circ)$ encodes exactly the atoms, every summand is a product of indicators of atoms; hence

$$\mathbf{1}[e \in q^\circ(G)] \;=\; \mathbf{1}\big[\hom_{D(q^\circ)}(G, e) \geq 1\big]\,.$$

Set $\mathcal{W}_{q^\circ} = \{D(q^\circ)\}$ (a single witness of order m); membership is decided by one anchored hom-count.

Step 2 (disjunction, DNF). If $q = \bigvee_\ell c_\ell$ with conjunctive clauses c_ℓ, then $\mathbf{1}[e \in q(G)] = 1 - \prod_\ell (1 - \mathbf{1}[e \in c_\ell(G)])$, a polynomial in the clause indicators, each of which is Step-1 decidable. The witness family is the union of the clause databases.

Step 3 (atom-level negation, inclusion–exclusion). Let $q = c \wedge \neg a_1 \wedge \cdots \wedge \neg a_k$ where c is conjunctive and the a_i are negated atoms (possibly with fresh existential variables, as in `pin`/`pni`). Then

$$\mathbf{1}[e \in q(G)] = \mathbf{1}[e \in c(G)] \cdot \prod_{i=1}^{k} \big(1 - \mathbf{1}[e \in (c \wedge a_i)(G)]\big).$$

Expanding the product,

$$\mathbf{1}[e \in q(G)] = \sum_{S \subseteq [k]} (-1)^{|S|}\, \mathbf{1}\big[e \in (c \wedge \bigwedge_{i \in S} a_i)(G)\big],$$

and each $(c \wedge \bigwedge_{i\in S} a_i)$ is again a positive conjunctive query whose canonical database D_S is a witness of order at most $m + \sum_i |\mathrm{var}(a_i)|$. Hence

$$\mathbf{1}[e \in q(G)] \;=\; \sum_{S \subseteq [k]} (-1)^{|S|}\, \mathbf{1}\big[\hom_{D_S}(G, e) \geq 1\big],$$

a finite integer combination of anchored hom-counts; negated atoms contribute through inclusion–exclusion as claimed. Combining Steps 2 and 3 handles arbitrary DNF with negation. The bound on witness order is $|\mathrm{var}(q)| + \sum_i |\mathrm{var}(a_i)|$, finite and independent of G. □

Remark. The theorem states a sufficient statistic for *decidability*; executors in practice approximate thresholded counts by smooth scores. The smooth statement used in the main text—order-k executors factor through order-k signatures—follows because every feature consumed by such an executor is a function of the signature by Definition 3, and signatures of order k determine signatures of order $\leq k$.

Proof of Theorem 2 (Separation)

Theorem 2. *There is an explicit, linear-time constructible pair of query worlds (G_1, q_1), (G_2, q_2) such that (i) every score-factored fuzzy pipeline outputs identical rankings for q_1 in G_1 and q_2 in G_2, while (ii) their ground truths differ and an order-2 witness aggregator separates them.*

Proof. Construction. Entities: anchors a, targets t_1, t_2, t_3, t_4, marker hubs z. Relations: r_a; ambiguous relations r_1, r_2; type markers τ_1, τ_2. In each *block* b: (a) r_a-edges from the anchor a_b to all four targets; (b) a hidden half $H_b \subset \{t_{b1}, ..., t_{b4}\}$ with $|H_b| = 2$, chosen uniformly; (c) each target in the r_1-half carries a τ_1-edge and each target in the r_2-half a τ_2-edge; (d) in *demonstrator* blocks only, the local r_1/r_2 edges from the anchor are observed. The query in a hidden block is $q_b = (a_b, r_a) \wedge (a_b, r_q)$ with $r_q \in \{r_1, r_2\}$; its ground truth is the two targets of the r_q-half, determined by the hidden edges—equivalently, by which targets carry τ_{r_q}-markers, since the construction ties the marker type to the half.

(i) Score-factored pipelines collide. Fix any executor whose projection returns per-entity scores and whose logical operators are pointwise functions of per-entity branch scores (products, probabilistic sums, complements, or any fixed continuous combinations thereof). In a hidden block the anchor has *no observed* r_q-edges; conditional on the block's local observations, the branch distribution over entities is a function of (a_b, r_a)'s local features only. By construction the local features of (a_b, r_a) are identical across the two worlds (both are "anchor with r_a-fanout four and unobserved r_q"), and they are identical for $r_q \in \{r_1, r_2\}$ within a world up to relabeling of relation symbols, which transferability (Definition 2) forces to act as an isomorphism. Hence the computed branch marginals, and therefore the final pointwise combinations and rankings, coincide across (G_1, q_1) and (G_2, q_2): the pipeline cannot be correct on both, because the ground truths differ (disjoint answer sets). Its expected MRR equals that of ranking four candidates of which two are relevant uniformly at random: $\frac{1}{4}(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}) \cdot 2 = \frac{2}{3}$ rounded to 0.67 (matches the empirical 0.673–0.703).

(ii) Order-2 aggregation separates. Define the aggregator that scores a target t in branch r_q by the indicator that t carries a τ_{r_q}-edge. This is a function of the order-2 signature: the pair (r_q, τ_{r_q}) co-occurrence is a two-node pattern count in the relation-interaction graph, and target markerhood is a one-hop anchored count. The aggregator's intersection then returns exactly the two relevant targets; MRR $= 1$. This is precisely the rule our order-2 gate variant implements (branch relation feature concatenated into the gate), and the empirical 1.000 matches. □

Remark (harmlessness in distribution). When the projection is trained to near-calibration on the query distribution, equal marginals co-occur with equal groundings, so the collision of (i) does not reduce accuracy—consistent with our paired-collision analysis on FB15k-237 (output Spearman 0.999, MRR ≈ 1). The separation bites exactly when local evidence is missing or shifted, which is the zero-shot regime of §6.3.

Proof of Theorem 3 (Learnable-Operator Characterization)

Theorem 3. *The class $\mathcal{O}_k$ of order-k operators satisfying transferability is exactly the class of branch-permutation-invariant functions computable from multisets of order-k branch signatures. Fixed continuous t-norms/t-conorms act pointwise on order-0 evidence and form a strict subclass of measure zero inside any smoothly parameterized order-≥ 1 family containing them.*

Proof. (*Structure of the class.*) An operator ω consumes the branch outputs of a plan prefix. Transferability (Definition 2) implies ω is invariant under entity and relation relabelings; since the plan's branch list is unordered (conjunction and disjunction are commutative in the query DAG), ω is additionally invariant under branch permutations. By definition of aggregation order, ω is a measurable function of the multiset $\{s_i, \sigma_i\}_{i \leq m}$ of branch scores and order-k signatures. Conversely, any branch-symmetric measurable function of those multisets defines an element of $\mathcal{O}_k$: it consumes no identifiers, and multiset arguments make the relabeling action an isomorphism, so equivariance holds.

(*Fixed t-norms are order-0 and strict.*) A continuous t-norm combination applied pointwise over entities depends on the branch score vectors only—order-0 evidence—hence lands in $\mathcal{O}_0 \subsetneq \mathcal{O}_k$ for $k \geq 1$; strictness is witnessed by Theorem 2's separator, which is not expressible by any pointwise combination.

(*Measure zero.*) Let $\{T_\theta\}_{\theta \in \Theta}$ be a smoothly parameterized family containing the product at θ^*, e.g., the Aczél–Alsina family with $\theta^* = 1$, and let the gate be a smooth function of parameters and evidence. The set of parameters for which the operator equals T_{θ^*} pointwise is the preimage of a point under a smooth map into a space of higher dimension; by Sard's theorem it has measure zero. (For the one-parameter Aczél–Alsina family the preimage of $\{\theta^*\}$ under a non-degenerate gate is a codimension-one submanifold.) □

Lemma 1 (Exact Degeneration)

Lemma 1. *Setting every* SQA *gate output to the constant c with $\sigma(c) = \frac{1-\theta_0}{3.5}\big|_{\theta_0=1}$, i.e. $\theta \equiv 1$, makes the gated intersection equal to the product t-norm for all inputs, and dually for union.*

Proof. With $\theta_i \equiv 1$ for all branches,

$$T_\Theta(s_1, \ldots, s_m) = \exp\Big[-\sum_i (-\ln s_i)\Big] = \prod_i \exp(\ln s_i) = \prod_i s_i,$$

using $\sum_i (-\ln s_i) = -\ln \prod_i s_i$ and continuity of exp; the dual statement for union follows by applying the identity to complements $1 - s_i$. Numerically we verify $\max |T_\Theta - \prod s_i| < 10^{-3}$ at the forced gate on random inputs. Note the exponent $1/\theta_1 = 1$ as well, so no branch asymmetry survives. □

Proposition (Transfer Continuity)

Proposition 1. *Define $d_{\hom}(G, G')$ as the ℓ_∞ discrepancy of anchored witness-count profiles over all patterns of order $\leq k$. For any order-k executor whose aggregation is L-Lipschitz in its inputs, $|F(G, q)(e) - F(G', q)(e')| \leq L\, d_{\hom}(G, G')$ for anchors e, e' corresponding under the profile.*

Proof sketch. The projection's messages are continuous functions of the anchored profiles (each hop composes counts along observed edges, Lipschitzly in the counts by the mean-value bound and the bounded degree of the plan). The order-k operator is L-Lipschitz in the branch scores and signatures by hypothesis. Chaining the two Lipschitz maps and taking the supremum over the finitely many order-$\leq k$ patterns yields the claim. The full multi-relational graphon formalism (existence of limits of anchored profiles, tightness) follows the classical bounded-degree convergence theory; we include it for completeness in the released technical note. □

Experimental Details Referred to in the Main Text

A. Hyperparameters

Projection: 2-layer conditional propagation per hop, hidden 64. Relation encoder: 3-layer typed MP over G_R, hidden 64. Gates: 2-layer MLP ($65 \to 64 \to 1$; order-2 variant $129 \to 64 \to 1$), $\theta \in (0.5, 4)$ via $0.5 + 3.5\sigma(\cdot)$. Training: AdamW, ranking loss (pretraining, lr 10^{-3}, 50k steps, batch 12) or official balanced logit-BCE (fine-tuning, lr $5{\times}10^{-4}$, 10k steps, batch 32). Total executor parameters ≈ 145k (gated).

B. Inference cost

Measured on one RTX PRO 6000, FB15k-237 ($|V|$=14,505, $|E|$=544k with inverses): fixed operators 21.3 it/s (batch 12, all types); gated 19.6 it/s; gated order-2 19.1 it/s—an 8–10% overhead over fixed operators, negligible against projection cost. Zero-shot executor fusion (router) doubles projection cost (two executors), $\sim 2\times$.

C. Gate statistics after official fine-tuning

Empirical θ on sampled `2i` queries concentrates in $[0.52, 0.57]$—systematically *softer* than the product point $\theta = 1$: the learned optimum relaxes conjunction strength rather than sharpening it, consistent with the large disjunction gains (`2u` 0.963) under fuzzy margins.

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